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RESEARCH ARTICLE

Multi-Modal Dyslexia Detection Model via SWIN Transformer With Closed-Form Continuous Time Networks

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ABSTRACT Dyslexia is a condition that reduces the cognitive abilities of the individuals. Dyslexic individuals (DI) face challenges in recognizing letters and words. Healthcare centers employ multiple modalities, including functional magnetic resonance imaging (MRI), Electroencephalogram (EEG), eye movements, and handwritten images for dyslexia detection (DD). A limited number of research studies employed multi-modalities for DD. The existing DD models demand substantial computational resources to identify dyslexia patterns. Ensemble learning (EL) approaches integrate the efficiency of multiple models to produce an optimal outcome. In this study, the authors developed an EL-based DD model for detecting dyslexia using MRI and EEG. An improved SWIN transformer was used to extract features from MRI images. A closed form continuous time neural networks (CFC) model was fine-tuned to detect crucial patterns from EEG data. The authors employed CatBoost and Dartbooster XGBoost (DXB) as base models for dyslexia prediction. An extremely randomized tree was used as meta-model to make a decision using base models outcomes. The model was developed based on two MRI datasets with an EEG dataset. The authors compared the proposed DD model's performance against single and EL-based DD models. The findings indicated a significant improvement in DD using the model. The proposed model delivered an exceptional performance by achieving 98.5% and 98.7% on MRI and EEG datasets. Educational institutions and healthcare centers can benefit from the proposed DD model by rendering better services to DI. Generalizing the proposed model in diverse datasets can improve its performance. The proposed DD model's performance can be enhanced using genome data.

INDEX TERMS Dyslexia, ensemble learning, SWIN Transformer, closed-form, continuous-time, neural networks, feature extraction, dimensionality reduction.

I. INTRODUCTION

Dyslexic individuals (DI) face challenges in reading and writing [1]. They may experience challenges in phonological processing, rapid naming, and working memory. The manifestation of dyslexia differs in each individual and may range from mild to severe [2]. Dyslexia is not associated with intelligence. DI typically have average or above-average intelligence. According to research findings [3], [4], multiple genes contribute to brain abnormalities and may lead to dyslexia. Neuroimaging studies revealed the variation in brain structure and functionalities of DIs [5]. These variations may affect the ability to comprehend phonological information and build effective neurological pathways for the reading process. Children are primarily affected by dyslexia. Dyslexic children (DC) typically experience poor self-esteem, frustration, and hostility [5].

An educational psychologist or expert may evaluate cognitive functioning, academic performance, and language abilities in a psychoeducational examination [5]. Standardized

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dyslexia and reading problem tests may be used to identify dyslexia. Reading, writing, spelling, phonological processing, and other skills are commonly assessed through standardized tests. These assessments evaluate decoding abilities, fluency, comprehension, and phonological awareness [6]. Dyslexia may be insufficiently represented in standardized tests. Decoding and understanding are typically assessed, whereas phonological processing, rapid naming, and working memory are commonly underestimated [6]. Standardized tests may benefit individuals with specific cultures or languages. In culturally varied groups, this bias can incorrectly interpret evaluation findings and misdiagnose dyslexia [7]. Administering and interpreting comprehensive psychoeducational examinations require time, resources, and expertise. Limited access to skilled personnel and evaluation tools may delay dyslexia diagnosis and treatment.

In Saudi Arabia, several studies examined the prevalence of dyslexia among individuals aged between 8 and 30 years. In study [8], the authors investigated the existence of dyslexia among 450 female students at King Saud University. The findings revealed that 2% of students had learning disorders. Alhifthy et al. [9] examined the prevalence of dyslexia in primary schools. They found a significant number of DCs in the Riyadh province of Saudi Arabia. Ewain et al. [10] highlighted the existence of learning disorders among students of Riyadh elementary schools. Similarly, the authors [11] identified the behavioral indicators of dyslexia among primary school students. The Saudi Arabian government initiated supportive measures to assist the educational centers in rendering effective learning environments to DC.

Screening tools are straightforward assessments to identify dyslexia risk. These techniques assess dyslexia signs, including phonological awareness, rapid naming, and decoding [12]. Healthcare and educational screenings may identify individuals who may require additional examination. In order to identify dyslexia, some schools conduct screening programs. Dyslexia screening techniques differ in sensitivity and specificity. These techniques may provide false positives or negatives, resulting in poor identification and unproductive referrals or missed interventions [13]. These techniques may not cover wider age groups. Standardized screening techniques may misjudge reading impairments in children and those with minimal reading experience, delaying dyslexia diagnosis [13]. DCs may overcome their challenges and succeed academically and professionally with assistance and intervention. Effective therapies emphasize phonological awareness, phonics, fluency, vocabulary, and multisensory training can support DC.

Machine learning (ML) algorithms can use behavioral tests, neuroimaging data, genetic markers, and educational transcripts to identify dyslexia patterns and characteristics [14]. These algorithms can discover minor brain anatomy and function variations, language processing, and reading skills. DL techniques can diagnose dyslexia effectively by examining screening, educational, and developmental data [15]. Individual variances and unique risk factor combinations allow DL techniques to deliver individualized evaluations and intervention suggestions. DL algorithms may track variations over time and evaluate intervention efficacy without human judgment or interpretation. Using DL algorithms, automated data analysis, and decision support systems may simplify dyslexia screening, minimize clinicians' and educators' efforts, and enhance dyslexia detection capabilities [15]. Through DL algorithms, researchers may discover novel biomarkers, prediction models, and dyslexia processes, improving diagnosis and treatment.

Dyslexia is a multifaceted disorder encompassing genetic, neurological, cognitive, linguistic, and environmental components [16]. Multi-modality data captures diverse causes of dyslexia, leading to an in-depth evaluation [17]. To better understand dyslexia risk factors, causes, and consequences, researchers and clinicians may combine data from numerous sources. Multi-modality data may reveal dyslexia biomarkers. Neuroimaging technologies, including MRI and diffusion tensor imaging, reveal dyslexic brain anatomy and function. Researchers may uncover dyslexia biomarkers for early detection, diagnosis, and intervention by integrating neuroimaging and other biological indicators [17]. Multi-modal DL algorithms have the potential to predict dyslexia incidence, severity, and progression. These algorithms may identify high-risk individuals, initiate measures, and monitor treatment outcomes. These models can identify dyslexia more effectively than single-modality techniques, which may overlook significant predictors or interactions.

EL methods enhance prediction accuracy by combining base model predictions [18]. Bagging, boosting, and stacking approaches improve dyslexia detection models' performance by integrating data from several sources or algorithms [18]. EL methods minimize model shortcomings and discrepancies to minimize model variance [19]. In addition, these approaches reduce overfitting and instability in dyslexia detection algorithms trained on small or noisy datasets by combining predictions from multiple models. Researchers have suggested numerous approaches for diagnosing dyslexia using data from reading/writing exams, web-based word games, eye tracking, MRI and EEG scans, videos, and images.

Recently, DL techniques have been applied for dyslexia identification due to their efficiency in delivering optimal outcomes. SWIN Transformer is a sophisticated computer vision-based DL model. It uses self-attention processes across shifting windows to gather local and global contextual information from images. It is particularly beneficial for activities requiring high-level interpretation of complicated visual patterns including MRI images. CFCs are neural networks developed to represent continuous-time instances. These networks are better at handling variable sampling rates and continuous-time dynamics than discrete-time models for time-series data including EEG signals. CFCs can be customized to identify crucial EEG patterns, making them ideal

for dyslexic signal analysis. CatBoost is a powerful gradient boosting technique that makes optimal use of category information. It is recognized for preventing overfitting and accurately handling massive datasets. Due to its effectiveness and simplicity, CatBoost is employed in classification and regression. DXB is a variation of the widely recognized XGBoost algorithm, which is renowned for its scalability and gradient boosting performance. It prevents overfitting using dropout methods to improve model generalization. It can be beneficial in challenging prediction tasks including dyslexia diagnosis. Extremely randomized trees generate several decision trees by selecting at random splits at each node rather than selecting the optimal split. These methods improve tree diversity, rendering the final model more resilient.

However, these models demand a substantial training to detect dyslexia with limited computational resources. Complexity and self-attention processes enable these models harder to comprehend, which may pose challenges in healthcare applications. Thus, the authors intended to build an EL-based DD model using multi-modality data. The contributions of this study are listed below:

1. An improved SWIN transformer-based feature extraction for extracting dyslexia patterns from MRI images.

The improved SWIN Transformer employ effective self-attention methods to extract the crucial features. Due to the lower computing demand, the proposed model can analyze MRI data rapidly in clinical settings.

2. An enhanced CFC-based feature extraction model to detect dyslexia using EEG data.

The enhanced CFC model better captures complicated brain activity temporal patterns and variations. This method improves dyslexia identification and classification by emphasizing on continuous-time dynamics and identifying small variations and abnormalities in EEG data.

3. An EL-based DD model using hyperparameter-optimized CatBoost, XGBoost, and an extremely randomized tree.

To effectively handle diverse data forms and complexity, the proposed modely employs multiple models. CatBoost and XGBoost can handle structured data containing multiple feature types, whereas ExtraTrees is noise- and variability-resistant. This combination enables the proposed model to identify dyslexia, which may entail different features and data sources.

The remaining part of the study is structured as follows: Section II discusses the characteristics of the existing DD models. The proposed methodology for detecting dyslexia is presented in Section III. Section IV offers the findings of the experimental analysis. The contributions and limitations of the proposed study are discussed in Section V. Finally, Section VI highlights the study findings and their implications in identifying dyslexia using multi-modality data.

II. RELATED WORKS

Researchers have suggested numerous DD models using single and multiple modalities. The majority of DD models use images, including handwritten samples, MRI, eye movement, and EEG patterns [20]. In addition, individual behaviors, electronic health records (EHR), and EEG data were used for DD. Dimensionality reduction techniques play a crucial role in extracting key patterns associated with dyslexia [20]. EEG data are generated by placing the electrodes on the scalp in order to measure the electric activity of the brain. Reduced event-related potentials and disruptions in oscillating brain rhythms may represent the neurobiological variations associated with dyslexia. Similarly, MRI computes the changes in blood flow and oxygenation in the brain. It provides neural activation patterns during cognitive tasks. Handwritten samples can be used to identify dyslexia. It reveals the specific patterns of individuals with dyslexia. For instance, letter reversal, inconsistent character formation, and poor spatial organization can assist educational institutions and healthcare centers in identifying DI. Graphometric analysis techniques are typically used to measure pen pressure and kinetic parameters of handwriting movements for DD. Nerušil et al. [21] suggested a DD model using eye movements. Vajs et al. [22] employed VGG16 network to detect dyslexia using eye movement data. Kariyawasam et al. [23] introduced a DL-based DD model.

Formoso et al. [24] utilized EEG data for detecting dyslexia. EEG enables researchers to investigate the dynamics of neural processes of individuals during cognitive activities. It can identify key neural markers of dyslexia and monitor the effectiveness of the intervention over time. Perera et al. [25] analyzed the EEG signals in order to determine the patterns of dyslexia. Usman and Muniyandi [26] proposed convolutional neural networks (CNN) based DD model using neuroimaging data. MRI visualizes brain activation patterns with optimal precision. It locates brain regions associated with reading and language processing. It can detect structural abnormalities in brain, including cortical thickness and variations in gray matter volume. Alqahtani et al. [27] discussed the recent tools and techniques for finding dyslexia. AlGhamdi [28] introduced an EL-based model for DD. In addition, the research studies [29], [30], [31], [32], [33], [34], [35], [36] highlight the significance of EEG and MRI in detecting dyslexia.

The existing DL-based DD model requires extensive training using labeled data to find dyslexia. The requirement of computational resources may reduce the possibility of DD implementation in healthcare centers. The trained model may not be generalized in a real-world DD environment. Physicians may face challenges in interpreting the findings of DL-based DD model. These knowledge gaps motivated the authors to employ the recent feature extraction techniques for developing DD with limited computational power.

III. MATERIALS AND METHODS

The complex nature of EEG and MRI data demand SWIN transformer and CFC models. However, these models require architectural and performance improvements to offer an exceptional outcome. CatBoost, XGBoost, and Extremely

randomized tree are widely applied classification models. To improve the performance, there is a demand for strategies to fine-tune these models key parameters.

An EL-based DD model was proposed to overcome the limitations of existing DD models. Figure 1 outlines the proposed methodology for detecting dyslexia using MRI and EEG data. A SWIN transformer-based feature extraction is suggested to extract intricate patterns of dyslexia from the MRI images. To extract features from EEG data, the authors suggested a CFC-based feature extraction model. They employed XGBoost and CatBoost as base models to predict dyslexia. Finally, an ExtraTree model was used as a meta model to make the final prediction using the base models' outcomes. The proposed DD model was generalized using the public datasets. Performance evaluation metrics were used to measure the capability of the proposed DD model in identifying dyslexic and non-dyslexic individuals.

A. DATA ACQUISITION

Three public repositories were used to obtain the MRI and EEG data. Dataset 1 [37] covers the annotated MRI images of 58 children. Dataset owners used stimuli of 60 words, and matched pseudo homophones were used to evaluate the participants. The participants were instructed to read the stimuli. The activity was repeated thrice with a short break of 3-5 minutes. MRI images of children were acquired using three Tesla MRI scanners. The images were classified into normal, isolated spelling deficits (ISD), and dyslexia.

Dataset 2 [38] included T1-weighted structural MRI images of 264 children. The dataset provider performed voxel-by-voxel correlation analysis to identify noisy and atypical images. They identified low-quality images using low similarity estimates. In order to extract different angles of MRI images from dataset 1, the authors built an image slicer. Each voxel in the 3D volumetric data contained signal information representing anatomical structures. The authors followed the axial slicing orientation to iterate the images to extract the image slices. At each iteration, an image was extracted from the MRI images. Each image was resized to 224×224 pixels. To train and generalize the proposed model, the authors combined the extracted images with Dataset 2 images.

The EEG dataset [39] encompassed the EEG and behavioral data of 130 children. The dataset provider conducted two experimental tests to evaluate the children. A photodiode-integrated monitor was employed to assess the children's activity. A 128-electrode Hydrocel Geodesic SensorNet connected to Net Amps 300 was used to obtain EEG data. EEG dataset provided raw EEG signal, Delta, Theta, Alpha 1, Alpha 2, and Beta 1 of dyslexic and non-dyslexic individuals. In addition, the authors employed data augmentation techniques [40], [41] to train the proposed model using MRI and EEG datasets. These datasets cover a wider range of individuals, supporting the authors to provide effective training. Table 1 presents the characteristics of the combined MRI dataset.

B. SWIN TRANSFORMER-BASED FEATURE EXTRACTION

The authors constructed a feature extraction (FE) model using a SWIN transformer as a backbone. Unlike traditional transformers, SWIN transformers split the images into non-overlapping shifted windows [42]. This process offered an efficient and scalable feature extraction. A hierarchical partitioning of the images was used to generate and integrate the image patches as a network. Using this approach, local and global features were extracted effectively. In addition, SWIN transformers enabled the adaptive patch generation based on the layer levels. The adaptive patch generation

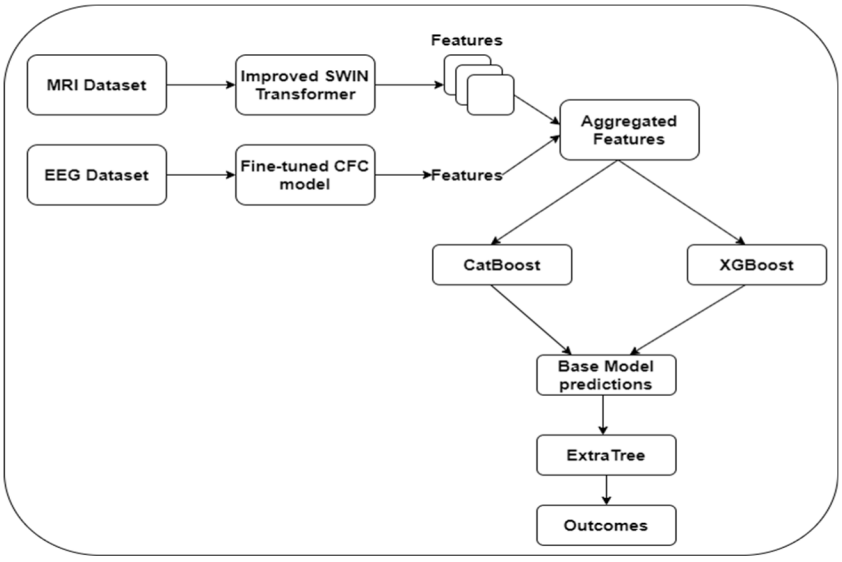


FIGURE 1. The proposed research methodology.

TABLE 1. Dataset characteristics.

Dataset		Classes	Train	Validation	Test
Combined Dataset	MRI	Dyslexia	742	247	247
		Normal	675	224	224
		Isolated	540	180	180
EEG Dataset		Dyslexia	339	113	115
		autism	309	103	101

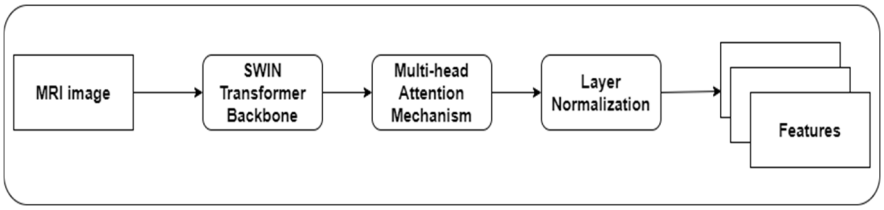


FIGURE 2. Improved SWIN transformer based FE.

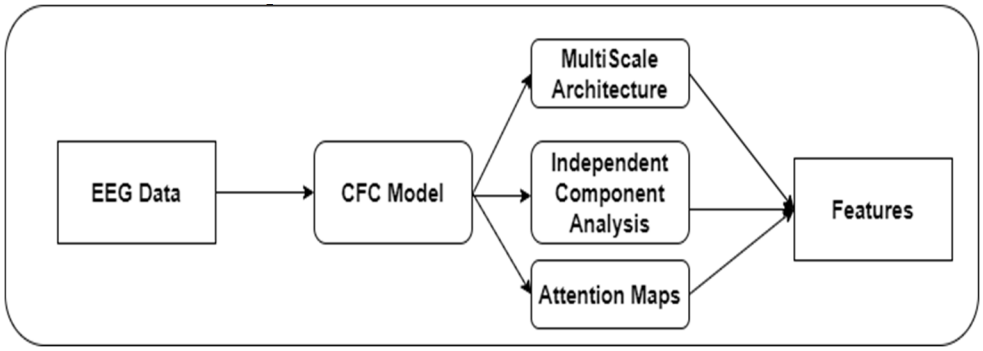


FIGURE 3. Improved CFC-based FE.

helped the transformer to extract features based on the region of interest. These features of the SWIN transformer motivated the authors to employ the SWIN transformer for the MRI feature extracton. The authors used the weights of the SWIN-L [42] transformer to construct the FE model. The initial layers were frozen in order to reduce the computing cost. Freezing the initial layers assisted the proposed model in generalizing on novel datasets. The authors integrated a multi-head attention mechanism to improve the FE capability. In addition, layer normalization was employed to extract the crucial features associated with dyslexia patterns. Figure 2 shows the FE methodology using MRI images.

Equation (1) shows the computational form of integrating attention weights to the SWIN transformer.

$$F = SWIN\ Transformer\ with\ Attention(MA\ (n, h)\ , LN) \tag{1}$$

where F is the features, MA is the multi-head attention mechanism, n is the number of heads, h is the head size, and LN is the layer normalization function to normalize the layers.

C. CLOSED FORM CONTINUOUS TIME NEURAL NETWORKS-BASED FE

EEG signal represents the electrical activity of the brain over time. It consists of lengthy time series data with temporal resolution. The complex temporal dynamics present cognitive processes, brain states, and neurological phenomena. The CFC model captures temporal dependencies in EEG data [43]. It offers a practical approach to extracting discriminative features of EEG signals. The authors employed the CFC model to identify dyslexic and non-dyslexic features from the EEG data. However, the CFC model operates at a fixed resolution based on the convolutional kernel size and the sampling rate. To overcome this limitation, the authors used multi-scale architecture to process data at multiple temporal resolutions to capture coarse-grained temporal patterns. The sensitive nature of the CFC model may degrade the FE process. Therefore, the authors applied independent component analysis to preprocess the EEG signals. They introduced attention maps to identify the key features of dyslexia. In addition, they integrated a flattened layer to flatten the features into a vector. Figure 3 highlights the improved CFC – based FE model.

Equation (2) presents the mathematical form of the CFC model.

$$F = CFC\ (C, FL, EEG) \tag{2}$$

where F is the feature, C is the convolutional layer, FL is the flattened layer, and EEG is the dataset.

D. CATBOOST

CatBoost is a gradient-boosting model that handles categorical features [44]. It implements an ordered boosting that incorporates categorical features into the training process. A histogram-based computation and distributed computing were used to reduce the computation time and memory consumption. CatBoost allows the developers to fine-tune the hyperparameters for the specific dataset and task requirements. The authors used CatBoost to predict dyslexia using the extracted features. To fine-tune the Catboost model, Optuna hyperparameter optimization framework was used [45]. It offers an efficient platform to optimize the gradient boosting models. The key idea is to find an optimal set of hyperparameters to improve the prediction accuracy. Using the Optuna algorithm, the authors applied the pruning process to stop the unpromising trail in the early stages. The objective function was evaluated using the sampled hyperparameters to generate a scalar value representing the CatBoost performance. Let P_1 and P_2 be the hyperparameters of the CatBoost model. Equations (3) and (4) highlight the parameters and objective functions to identify the optimal hyperparameters.

$$H = \{P_1 \leftarrow [0, 1], P_2 \leftarrow [0, 1]\} \tag{3}$$

$$P^* = argmax_{P_{1,2}} \leftarrow f(x) \tag{4}$$

where H is the hyperparameter, P_1 , P_2 and P^* is the optimal hyperparameter, and $f(x)$ is the objective function.

E. XGBOOST

XGBoost is a robust gradient-boosting library based on an efficient and scalable tree-boosting approach [46]. The XGBoost model aims to optimize a loss function by integrating weak learners sequentially. The excellent predictive performance, flexibility, and scalability of the XGBoost model motivated the authors to employ it in this study for feature extraction. To prevent overfitting and improve the generalization, the authors employed the Dartbooster-based XGBoost (DXB) model. During training, the Dartbooster technique applies dropout regularization to prune trees with specific probabilities. It uses stochasticity in the gradient-boosting process to improve the generalizability of the proposed DD model. The authors employed Bayesian optimization Hyperband (BOHB) hyperparameter optimization to fine-tune the DXB model [47]. BOHB is a multi-fidelity optimization that depends on budget. It relies on Hyperband optimization to determine the critical parameters of the DXB model. It uses a parallelization strategy to minimize the computation time and generate optimal results. To improve the generalization, Equation (5) shows the mathematical form of the DXB-based prediction of dyslexia.

$$Output = DXB\,(features, BOHB\,(obj, b)) \tag{5}$$

where *output* is the base model predictions of *DXB*, *obj* is the objective function, and b is the budget for optimizing the DXB model using BOHB optimization.

F. EXTRATREE-BASED DD

ExtraTree follows the concept of Random Forest and applies additional randomness in the tree-building process [48]. It aggregates the outcomes of the multiple decision trees to generate an outcome. Random feature selection and random thresholds were used to split the features. In this study, the authors employed ExtraTree as a meta-model to predict dyslexia using the results of CatBoost and DXB models. A random subset of features was used to train the tree. To fine-tune the hyperparameters, including the number of trees, maximum depth of trees, minimum number of samples, and maximum number of features, the authors used the Optuna optimization algorithm. Optuna models the objective function using a tree-structured Parzen estimator. It determines the value of the objective function and uncertainty for hyperparameter configurations. In addition, the acquisition function identifies the key parameter configurations to improve the model's performance. The authors employed SHAP to interpret the model's predictions. Aggregating SHAP values into the proposed DD model facilitates the assessment of feature importance and identifies the key patterns of dyslexia. Equations (6) and (7) present the hyperparameter optimization and integration of SHAP values.

$$output = ExtraTrees\,(C\,(output)\,, D\,(output)\,, Optuna) \tag{6}$$

where *output* is the findings of ExtraTrees, C is the CatBoost outcomes, D is the DXB model's outcomes, and *Optuna* is the hyperparameter optimization.

$$SHAP_values = SHAP.Explainer\,(ExtraTrees\,(outcomes)) \tag{7}$$

where *SHAP* values indicate the interpretable outcomes using the *ExtraTrees* (*outcomes*) and *SHAP.Explainer* is used to interpret the findings of the *ExtraTrees*() function.

G. EXPERIMENT VALIDATION

The authors used evaluation metrics, including accuracy, precision, recall, F1-score, Mathew correlation coefficient (MCC), and Cohen's Kappa to measure the capability of the proposed DD model in detecting dyslexia. Equations (8) and (9) present the mathematical form of evaluation metrics.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{8}$$

$$Cohen's\ Kappa = \frac{P_o - P_e}{1 - P_e} \tag{9}$$

where TP is the true positive, representing the number of individuals correctly identified as dyslexic, TN is the true negative, indicating the number of non-dyslexic individuals correctly identified as non-dyslexic, FP is the false positive, representing the number of non-dyslexic individuals incorrectly identified as dyslexic, FN is the false negative,

highlighting the number of DI incorrectly identified as non-dyslexic, P_o is the proportion of instances where raters agree, and P_e is the proportion of instances where agreement is expected by chance.

The area under the receiver operating characteristics (AU-ROC) was used to measure the proposed DD model's ability to distinguish between the positive and negative classes. The area under the precision-recall curve (AU-PRC) presents the trade-off between precision and recall across multiple decision thresholds. In addition, the authors used standard deviation (SD) and confidence interval (CI) to quantify uncertainty and assess the variability of the study findings.

IV. RESULTS

The recommended DD model was developed using Windows 11, Intel i9, 16 GB RAM, and NVIDIA GTX 3050X environment. In order to measure the model's performance, the datasets were divided into a train set (60%), a validation set (20%), and a test set (20%). Python 3.8.3 was used to build the model. Keras 2.11.0, TensorFlow 2.9.0, Theano 1.0.5, and PyTorch 2.0 were used to implement the proposed feature extraction, base, and meta models. The model was trained using batch sizes of 36 and 42 for MRI and EEG datasets, respectively. Using the validation set, the authors improved the model's detection capability. Early stopping strategies were applied based on the validation loss. In order to enhance the performance of the proposed DD model, the authors employed quantization-aware training (QAT) [49]. QAT integrates simulated quantization steps, including scaling and routing, into the training process. It supports the proposed model to reduce the computational loss. It assists the proposed DD model in maintaining a trade-off between computational resources and optimal accuracy. The authors trained the model upto 47 epochs for the combined MRI and 54 epochs for EEG datasets, respectively. The pre-trained CNN models, including MobileNet V3, SqueezeNet V1.0, and EfficientNet B7, were used for the comparative analysis. In addition, the performance of recent DD models was compared against the proposed DD model. The experimental settings are presented in Table 2. The performance of the recommended DD model during the training and validation is highlighted in Figures 4 and 5. It is evident that the recommended DD model achieved optimal accuracy with minimal computational loss on the validation set. The authors applied the loss function in order to evaluate the potential of the proposed model in generating optimal results with minimal error. Reduced computational loss speeds up proposed model predictions for massive datasets or real-time applications. Less processing power and memory may lead to cost savings, particularly in resource-constrained environments, including healthcare facilities and educational institutions. The minimal computational loss renders the technology accessible to smaller medical facilities and educational institutions without significant computational resources.

TABLE 2. Experimental settings.

Parameters		Values
SWIN Transformer Model		
	Image size	224 × 224
	Patch size	4 × 4
	Number of heads	4
	Embedding dimension	81
	Window size	4
	In_channels	3
CFC Model		
	Convolution Layers	3
	Pooling layers	2
	Batch normalization	2
	Stride	1,1
	Groups	12
CatBoost		
	Iterations	1000
	Learning rate	0.1
	Random seed	42
	Depth	6
XGBoost		
	Maximum depth	6
	Number of boost round	100
	Eta	0.1
	Objective	Binary: logistic
ExtraTrees		
	Number of estimators	100
	Maximum depth	7
	Minimum samples split	3
	Random State	42

The findings of the performance analysis are highlighted in Table 3. The EL-based DD model captured the complex heterogeneity of dyslexia and offered a significant performance. In addition, optimal MCC, Kappa, precision, recall, and F1-score outlined the model's ability to distinguish between dyslexic and non-dyslexic individuals. The suggested Catboost and DXB models contributed to the exceptional performance of the proposed model.

The authors compared the performance of the proposed model with existing models. Table 4 demonstrates that the proposed DD model obtains an accuracy of 98.5%. The model outperformed the existing models in terms of accuracy, precision, recall, F1-score, MCC, and Kappa, indicating its superior capability in detecting dyslexia using MRI images.

The authors generalized the model on Dataset 1. Table 5 reveals the findings of generalization. The findings demonstrate significant generalizability with no signs of overfitting issues. The consistent performance across the data samples indicates the model's potential as a reliable tool for early diagnosis of dyslexia. It underscores that the proposed model can be deployed in the resource-constrained environment. The outcomes of the comparative analysis using the EEG dataset are reported in Table 6. The use of FE techniques and hyperparameter optimization strategies contributed to the superior performance of the recommended model. The recommended EL approach enabled the integration of critical information from EEG data, leading to optimal accuracy. Table 7 reveals the outcomes of reliability and uncertainty analysis. An AU-ROC of 96.0 for the combined MRI dataset,

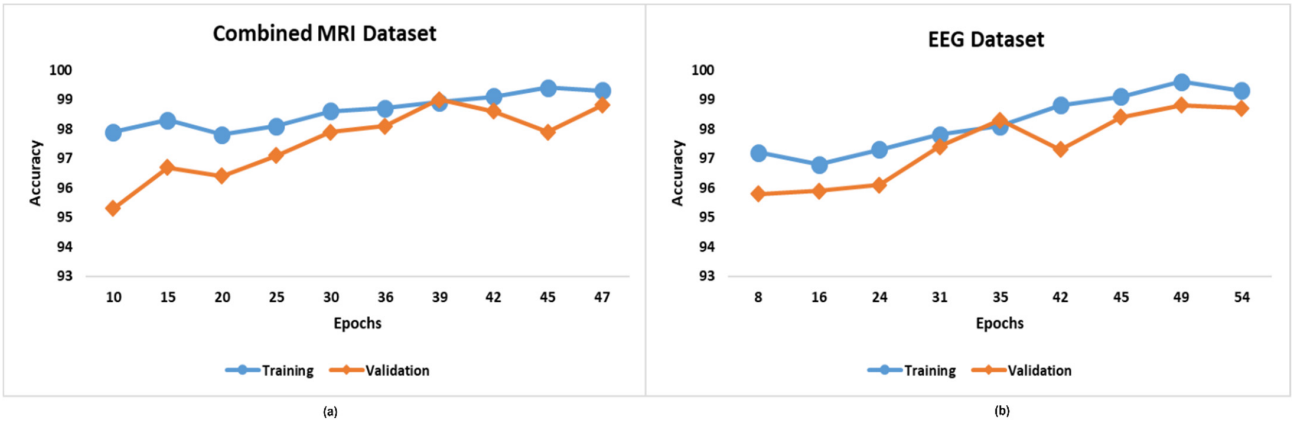


FIGURE 4. Accuracy – Training and validation (a) Combined MRI dataset (b) EEG dataset.

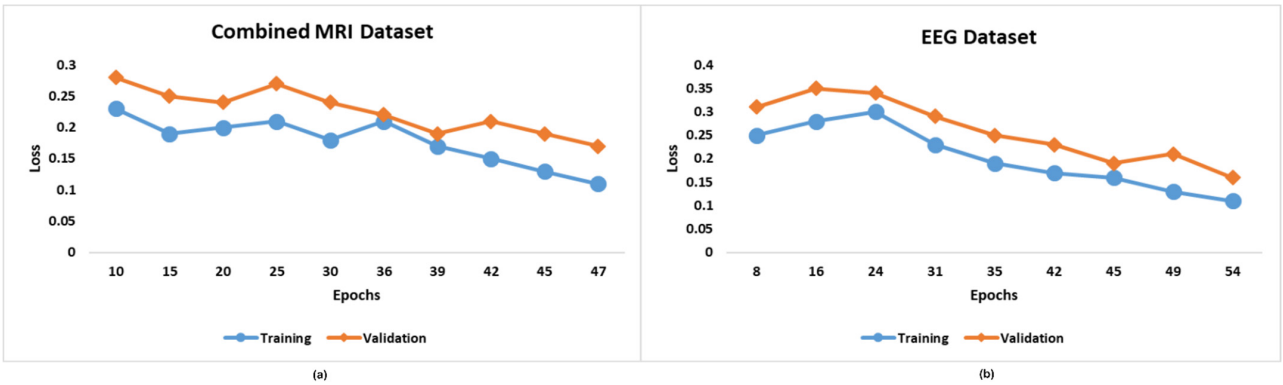


FIGURE 5. Loss – Training and validation (a) Combined MRI dataset (b) EEG dataset.

97.0 for Dataset 1, and 95.0 for the EEG dataset represents that the proposed model achieved high true positive rates by maintaining low false positive rates. It demonstrated the robust discriminative power of the proposed model. Likewise, an AU-PRC score of 92.0, 94.0, and 94.0 reflected the model's ability to identify dyslexia with fewer false positives. The high AU-ROC and AU-PRC scores of the proposed model have significant implications for educational interventions and clinical practices. Compared to the current models, the proposed model obtained SD of 0.0003, 0.0004, and 0.0005 with CI of [95.9 – 96.4], [96.5-96.9], and [96.5 – 97.1], highlighting the reliability of the findings. The proposed DD model utilized limited parameters and FLOPs with less computation time. It can be deployed in real-world scenarios, enabling rapid and accurate dyslexia screening.

V. DISCUSSIONS

An EL-based DD model was developed in this study using MRI and EEG datasets. A fine-tuned SWIN transformer backbone-based feature extraction was used to generate the crucial features of dyslexic and non-dyslexic images. The CFC model was used to extract features from EEG data. The CFC mode's performance was improved using the adaptive weighted feature importance analysis technique. Two base models, including CatBoost and XGBoost models, were used to predict dyslexia using the extracted features. Lastly, the authors employed ExtraTrees as a meta-model to make a final prediction using the base model' predictions. The experimental settings are described in Table 2. The performance of the EL-based DD model was outlined in Tables 3-7. In addition, Figures 4–5 highlight the effectiveness of the proposed DD model. Data augmentation supported the authors to balance the dataset with an equal representation of dyslexic and non-dyslexic individuals. It assisted the proposed model in maintaining consistent performance across diverse datasets. The AU-ROC of 96.0 for the combined MRI dataset, 97.0 for dataset 1, and 95.0 for the EEG dataset, and AU-PRC of 92.0 for the MRI dataset, 94.0 for the dataset 1, and 94.0 for the EEG dataset indicated the robustness of the proposed model in differentiating dyslexia and non-dyslexia patterns. The outcomes of uncertainty analysis indicated the reliability of the proposed DD model in finding dyslexia. The proposed model can be employed to conduct large-scale screening programs for dyslexia using multi-modality data. It can be generalized on diverse MRI and EEG datasets.

The existing pre-trained CNN models require a large amount of samples to detect dyslexia using the MRI images. The MobileNet V3 model obtained an average accuracy of 97.2% with substantial computational resources. EfficientNet B7 and SqueezeNet V1.0 models showed a better performance in detecting dyslexia. These models generated a significant set of features for DD. However, they demanded

additional memory and computational power that may reduce their performance in the resource-constrained environment. Fine-tuning the EfficientNet B7 and SqueezeNet V1.0 models requires considerable annotated data on dyslexia to generalize on diverse datasets. In contrast, the proposed DD model demanded few parameters and FLOPs for DD using the MRI images.

The existing models employ a pre-trained CNN model to detect dyslexia using MRI images. The lack of feature importance analysis reduces its performance. The reliability of these models is lower compared to the proposed model. Similarly, the models based on the EEG dataset require substantial computational power to identify dyslexia. These models suffer from vanishing and exploding gradient problems. In addition, they have a fixed internal memory, leading to the loss of relevant context and hindering the model's capability to retain crucial data. Additionally, the proposed model facilitates the model interpretability using the SHAP values. For instance, a SHAP value of -0.02 significantly lowers the dyslexia risk. Similarly, the SHAP value of $+0.06$ indicates the risk of dyslexia. Using the recommended SHAP values, physicians can comprehend the proposed model's decision.

The proposed EL-based DD model can reliably differentiate dyslexic and non-dyslexic individuals. The model can offer a compelling diagnosis of dyslexia in clinical settings. Using the proposed model, educational institutions and healthcare centers can identify dyslexia in the early stages. The model can uncover unique biomarkers associated with dyslexia. It can assist physicians in comprehending the neurological basis of dyslexia. By identifying the neural mechanisms of dyslexia, healthcare centers can offer personalized treatment to DIs. The multi-modality-based DD model can reduce the complexities associated with traditional dyslexia assessment techniques.

The integration of CatBoost and XGBoost models produced greater precision and resilience. The recommended EL model can identify dyslexia in children, enabling educational institutions to implement individualized early intervention programs. In order to provide individualized rehabilitation programs, rehabilitation institutions can employ the proposed DD model. By precisely identifying DC, education institutions can establish customized learning strategies based on their strengths and weaknesses. DC's academic performance and well-being can be improved using customized learning strategies. To optimize resource allocation, the proposed

TABLE 3. Findings of performance validation.

Dataset	Classes	Accuracy	Precision	Recall	F1-Score	MCC	Kappa
Combined MRI Dataset	Dyslexia	99.5	99.2	99.1	99.1	97.2	97.8
	Normal	99.3	99.4	98.8	99.1	97.9	98.4
	ISD	99.7	99.6	99.1	99.3	96.8	97.7
EEG Dataset	Dyslexia	99.5	99.3	98.7	99.0	97.1	97.9
	autism	99.6	98.9	98.6	98.7	97.5	98.1

TABLE 4. Comparative analysis – combined MRI dataset.

Models	Accuracy	Precision	Recall	F1-Score	MCC	Kappa
Proposed Model	99.3	99.2	98.9	99.0	97.1	97.4
Harismithaa and Sadasivam [29]	97.2	96.1	96.7	96.4	94.3	95.7
MobileNet V3	97.1	95.4	95.6	95.5	91.5	92.6
EfficientNet B7	96.8	96.7	96.8	96.7	92.7	93.4
Silva et al. [30]	94.7	94.5	94.4	94.4	91.6	93.2
SqueezeNet V1.0	96.1	96.4	96.7	96.5	90.8	92.4

TABLE 5. Comparative analysis – dataset 1.

Models	Accuracy	Precision	Recall	F1-Score	MCC	Kappa
Proposed Model	99.4	99.0	98.6	98.8	97.4	97.8
MobileNet V3	96.7	97.4	97.8	97.6	95.1	95.5
EfficientNet B7	97.1	96.8	97.2	97.0	94.7	95.8
SqueezeNet V1.0	96.9	97.3	97.5	97.4	95.4	96.1

TABLE 6. Comparative analysis – EEG dataset.

Models	Accuracy	Precision	Recall	F1-Score	MCC	Kappa
Proposed Model	99.7	99.3	98.9	99.1	96.9	97.6
Bi-LSTM Model	97.1	96.6	96.8	96.7	94.3	93.6
Martinez-Murcia et al. [31]	90.2	90.5	89.6	90.0	89.5	88.6
Parmar et al. [32]	96.7	95.8	94.7	95.2	91.5	93.7
Seshadri et al. [33]	97.5	97.1	97.3	97.2	90.8	96.5
Bi-GRU Model	94.8	95.1	95.7	95.4	91.5	90.8

TABLE 7. Findings of reliability and uncertainty analysis.

Dataset	Models	AU-ROC	AU-PRC	SD	CI	Parameters	Flops	Time (seconds)
Combined MRI Dataset	Proposed Model	96.0	92.0	0.0003	[95.9-96.4]	25	29	0.191
	Harismithaa and Sadasivam [29]	92.0	93.0	0.0005	[95.6 – 96.3]	36	39	0.357
	MobileNet V3	92.0	90.0	0.0004	[95.1-96.2]	42	38	0.489
	EfficientNet B7	93.0	88.0	0.0003	[95.1-96.4]	45	37	0.312
	Silva et al. [30]	90.0	89.0	0.0004	[96.2 – 96.7]	42	29	0.453
	SqueezeNet V1.0	91.0	87.0	0.0005	[95.8-96.5]	35	42	0.512
Dataset 1	Proposed Model	97.0	94.0	0.0004	[96.5-96.9]	21	20	0.106
	MobileNet V3	93.0	91.0	0.0005	[95.8-96.1]	31	34	0.247
	EfficientNet B7	93.0	92.0	0.0004	[96.1-96.7]	37	36	0.254
	SqueezeNet V1.0	92.0	93.0	0.0004	[95.8-96.6]	39	39	0.234
EEG Dataset	Proposed Model	95.0	94.0	0.0005	[96.5 – 97.1]	30	28	0.212
	Bi-LSTM Model	91.0	90.0	0.0004	[95.7-96.5]	44	39	0.418
	Martinez-Murcia et al. [31]	91.0	90.0	0.0004	[95.0-95.8]	51	42	0.504
	Parmar et al. [32]	90.0	91.0	0.0003	[95.4-96.1]	54	44	0.589
	Seshadri et al. [33]	92.0	91.0	0.0004	[95.1-96.7]	47	51	0.631
	Bi-GRU Model	93.0	89.0	0.0003	[95.1-95.9]	56	49	0.723

model can support teachers to prioritize students most likely to benefit from intervention initiatives. These programs may improve DC reading and writing skills and quality of life by targeting cognitive and linguistic skills. Researchers can use the proposed model to understand dyslexia's neurological underpinnings and improve diagnosis by examining massive dyslexic datasets.During the training phase, the authors faced challenges, including dataset heterogeneity, dataset size limitations, and model interpretability. In order to address the dataset heterogeneity, the authors improved the SWIN transformer to extract the features by focusing on specific brain regions associated with reading-related tasks. Similarly, the CFC model was fine-tuned to identify the key features of dyslexia. The authors applied the data augmentation techniques to overcome the sample size limitations. They employed SHAP values in the proposed DD model to facilitate the interpretation of the findings. By leveraging SHAP values, clinicians can gain insights into feature significance, model behavior, and decision-making processes. The proposed model facilitates the SHAP values of the features associated with each prediction. Thus, clinicians can comprehend the decision-making process of the proposed model.

Due to fundamental variations, EEG and MRI data heterogeneity makes feature capture challenging. MRI offers high-resolution anatomical images of the brain's structure, while EEG data shows brain electrical activity in real time. EEG data need temporal analysis to capture brainwave patterns, whereas MRI data require spatial analysis to find structural anomalies, complicating feature extraction. CFC models should be fine-tuned to extract significant patterns from high-frequency, low-amplitude EEG data. Similarly, MRI features are huge, high-dimensional image, requiring multi-head attention mechanisms and SWIN model to capture tiny anatomical variations. The authors encountered challenges in integrating these features into a unified model due to the fact that MRI spatial resolution is higher than EEG temporal resolution, which might result in the loss of crucial patterns. The authors analyzed the MRI and EEG datasets and mitigated biases. They performed a thorough analysis of features in order to identify the presence of significant biases that influence the findings. High accuracy and high variance can occur from overfitted CatBoost and XGBoost models. Without sufficient trees, Extremely Randomized Trees may generate bias. Achieving robustness without losing precision is challenging. As models undergo modifications and ensembles are constructed, CatBoost and XGBoost feature significance measures become less visible, making it harder to extract relevant insights. To overcome these challenges, the authors used the SHAP values to facilitate the model interpretability. They applied QAT in order to reduce the memory usage and accelerate inference in shorter duration. The inherent pruning strategies of the Extremely randomized tree decreases the computational load without affecting the mode's performance.

Encryption, anonymization, and secure storage practices were followed to protect individuals' sensitive attributes. Ethical evaluation and stakeholder interaction are essential to balance the proposed DD capability with ethical responsibilities. Establishing an ethical committee to monitor the deployment of the proposed model assists in addressing ethical issues. The members of this committee can offer guidance on bias prevention, privacy, and AI ethics.

Engaging with educators, physicians, patients, and their families during the implementation process ensures that the proposed system meets their requirements and ideals. By establishing a continuous monitoring practice, educational institutions and healthcare centers can balance the mode's effectiveness and ethical standards.

The recommended CFC and SWIN transformer enable the seamless integration of the proposed model with advanced applications. The proposed DD model can be implemented with the existing diagnostic applications in order to assess individuals. It can be integrated as a software plugin that facilitates smooth data exchange and analysis. It can enhance

diagnostic accuracy through data-driven and objective analysis. By conducting workshops and providing user manuals, clinicians can use the proposed model and understand the model's outcomes. Educational institutions and healthcare centers can deploy the proposed DD model with limited computational resources. The implementation costs can be reduced by deploying the model on cloud-based platforms. Cloud-based platforms can improve the mode's scalability and accessibility. An intuitive user interface can be developed that allows educators and clinicians to interact with the proposed model and visualize the results.

VI. CONCLUSION

In this study, the authors presented a novel DD model leveraging the EL approach and multi-modal data. They generalized the proposed DD model on MRI and EEG datasets. The experimental analysis outcomes demonstrated the effectiveness and robustness of the proposed model in identifying DI. By integrating an improved SWIN transformer and CFC model, the model achieved superior performance compared to the existing individual and EL-based DD models. The integration of MRI and EEG data provided crucial patterns associated with brain structure and function, enriching the feature representation for DD. In addition, introducing SHAP values has facilitated model interoperability by offering insights into decision-making processes. The proposed model can deliver promising results in clinical settings and educational interventions. Using this model, educational institutions and healthcare centers can improve DI's academic outcomes and quality of life. However, the proposed model faced challenges in extracting crucial patterns of dyslexia from multi-modal data. EEG and MRI data have diverse forms and features, causing challenges in the process of feature extraction. MRI records anatomy, whereas EEG records brainwaves. Aggregating unique data from the two modalities is necessary to integrate these distinct modalities into a unified model. EEG signal noise and distortions may obscure dyslexia-related signals. Noise in EEG data makes it hard to distinguish valuable signals. This hinders the extraction of reliable dyslexia diagnostic characteristics. A substantial training was required in order to maintain the tradeoff between optimal accuracy and computational complexities. The recommended optimization techniques addressed the limitations and assisted the proposed model in delivering an exceptional result. Furthermore, validating the proposed model on the independent datasets can enhance its interoperability. The proposed model highly depends on MRI and EEG data availability and quality. Data quality and consistency may influence model performance. Due to ethical, economical, and logistical restrictions, obtaining high-quality MRI and EEG data for dyslexia identification may be challenging. Due to the model's dependence on hyperparameter optimization for optimal performance, the efficacy of the model is hyperparameter-sensitive. Suboptimal hyperparameters may require frequent modification and testing. Although the model includes MRI and EEG data, incorporating genome or eye-tracking data may be complicated and require significant modifications. Without integrating these data sources, the model may not properly depict dyslexia's multifactorial characteristics. Integrating genetic data into dyslexia detection studies requires multiple strategic steps and may provide major advantages. Genome data may reveal dyslexia-related gene differentials. Imaging and EEG data are combined with these characteristics. Combining genetic data with additional characteristics creates a complete model input set with feature-level fusion. The use of genetic data in dyslexia research is significant. The approach may improve generalization accuracy using genetic data. This may improve predictions and comprehension of the dyslexia condition. Genetically targeted therapies may be developed through personalized medicine.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest to report regarding the present study.

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